FACULTY OF COMPUTING

Semester 01 2024/2025

Subject	:	**Database (SECD2523)**
Section	:	
Task	:	**Phase 3 (P3) – Database Logical Design & SQL**
Due	:	**WEEK 14 (10 January 2025) – draft (as a weekly task) (2.5%)** **WEEK 16 (24 January 2025) (20%)**

P3 Guidelines

No	Task	Deliverables (Items in Database)
1.	• Transform the conceptual ERD in P2 into logical ERD. (e.g.: by removing non-relational features e.g.: *:* relationships, complex relationships, etc.) • Derive relations schema from the logical ERD produced above. • Perform normalization up till BCNF (*if-any)* to these relations. • Draw the final logical ERD. • Update the data dictionary based on the normalized relations produced from above. • Validate logical ERD with the system's transaction requirements using interface design.	A report containing the following: • Logical ERD (global data model) • Relational database schemas (normalized table) • Data dictionary for the normalized logical design. • Interface design (mapped to the proposed SQL statement) • Proposed SQL statement (refer to rubric) Demonstration: • Individual demonstration for SQL statement.

Database Logical Design

Prepare a Database conceptual design as below:

1.0 Introduction
2.0 Overview of project
3.0 Database conceptual design
  3.1 Updated business rule
  3.2 Conceptual ERD
4.0 DB logical design
  4.1 Logical ERD
  4.2 Updated Data Dictionary
  4.3 Normalization
5.0 Relational DB Schemas (after normalization)
6.0 SQL Statements (DDL & DML)
7.0 Summary

P3 Rubrics

1. **Reporting (10%)**
   A. **Overall (2%)**

Criteria	High (4)	Average (3)	Low (2)	Unsatisfied (1)
Document the whole process of designing the ERD.	Documented **ALL** reports on the construction of the database design. Shows **FULL** understanding of the requirements based on the case study.	Documented **MAJOR** reports on the construction of the database design. Shows a **GOOD** understanding of the requirements based on the case study.	Documented **MINIMAL** reports on the construction of the database design. Shows **MINIMAL** understanding of the requirements based on the case study.	**LITTLE** evidence that the documents are based on the given case study.
SQL documentation	The student provides **THOROUGH** documentation for their SQL code, including comments explaining the purpose of the code, any complex logic, and the structure of the database.	The student provides **ADEQUATE** documentation for their SQL code, but it may lack some details or clarity in explaining complex logic.	The student provides **MINIMAL** documentation for their SQL code, making it challenging to understand the purpose and logic.	The student provides **NO** documentation for their SQL code, making it nearly impossible to understand the purpose and logic.

B. Database Logical Design – ERD & Normalization (6%)

Criteria	High (4)	Average (3)	Low (2)	Unsatisfied (1)
Follow the instructions to draw the logical ERD and derive the relations	**COMPLETE** understanding of how to draw ERD and derive relations.	Exhibits a **GOOD** understanding of how to draw ERD and derive relations.	Exhibits **MINIMAL** understanding of how to draw ERD and derive relations.	**UNABLE** to follow the instructions on how to draw ERD and fails to derive relations.
Identifies all the basic concepts in normalization (relational schema)	**COMPLETE** understanding of the basic concepts in ERD.	Exhibit a **GOOD** understanding of the basic concepts in ERD.	Exhibit a **MINIMAL** understanding of the basic concepts in ERD.	**UNABLE** to show understanding of the basic concepts of ERD.
Represents the dependency diagram (and relational schema) based on the given case study	**COMPLETELY** and **ACCURATELY** use of names, definitions in entities, and attributes - table name is appropriately used regarding its data elements.	Some **MINOR** inaccurate use of names, definitions in entities, and attributes - **FEW** table names are not clear regarding its data elements.	**INACCURATE** the use of names, definitions in entities, and attributes - table name DID NOT correlate with its data elements.	**WRONG** the use of names, and definitions in entities and attributes. Shows NO understanding in identifying the tables based on the case study.
Differentiates the keys in the dependency diagram	**COMPLETE** understanding of the dependencies and normal-form levels.	Exhibit a **GOOD** understanding of the dependencies and normal-form levels.	Exhibit a **MINIMAL** understanding of the dependencies and normal-form levels.	**UNABLE** to differentiate the dependencies and normal-form levels.
Label the necessary transitive and partial dependencies	Correctly choose **ALL** of the PKs and FKs, about the naming convention	Correctly choose **MOST** of the PKs and FKs, regarding the naming convention.	**INCORRECTLY** choose the PKs and FKs.	**WRONGLY** choose the PKs and FKs.

Criteria	High (4)	Average (3)	Low (2)	Unsatisfied (1)
Illustrates the normalization steps - from 1NF --> 2 NF --> 3NF -->BCNF (if applicable)	A **COMPLETE** populated the tables with **CORRECT** data elements as outlined in the ERD design.	Populated the tables with **ALL** data elements as outlined in the ERD design - with **MINIMAL** errors.	Populated the tables with **MINIMAL** data elements as outlined in the ERD design - with **MINIMAL** errors.	Populated the tables with **MINIMAL** data elements as outlined in the ERD design - with **NUMEROUS** errors.

C. Interface (2%)

Criteria	High (4)	Average (3)	Low (2)	Unsatisfied (1)
Appropriateness to target user	Prototype uses terms and symbols **APPROPRIATE** to target user. Look and feel **APPROPRIATE** to the target user. Usability **APPROPRIATE** to target user.	Prototype designed for a **WELL-DEFINED** target user. Usability is **APPROPRIATE** to the target user.	Prototype designed **WITHOUT** well-defined target user in mind. **SOME** usability is **NOT** appropriate for the target user.	Prototype designed **WITHOUT** well-defined target user in mind. **MOST** usability is **NOT** appropriate for the target user.
Prototype's Interaction Usability	The user interface uses a **SIMPLE** and **NATURAL** interaction style. **INTUITIVE, EASY to learn and use.**	The user interface uses a **SIMPLE** interaction style. **EASY to use once you learn it.**	**SOME** user interface is **COMPLICATED** and **HARD** to use.	**HARD** to figure out how to even get started using the prototype.

Criteria	High (4)	Average (3)	Low (2)	Unsatisfied (1)
Mapping SQL statements to the appropriate functions	Demonstrates an **EXCEPTIONAL** ability to map SQL statements to appropriate functions, showing **MASTERY** in this skill. **CONSISTENTLY** and **INNOVATIVELY** maps SQL statements to appropriate functions, optimizing performance and accuracy.	**CONSISTENTLY** and **ACCURATELY** maps SQL statements to the appropriate functions. **CONSISTENTLY** maps SQL statements to appropriate functions.	**ABLE** to map SQL statements to **APPROPRIATE** functions but still needs **IMPROVEMENT**. **INCONSISTENTLY** maps SQL statements to appropriate functions.	Demonstrates a **BASIC** understanding of SQL statements and functions but **STRUGGLES** to map them effectively. **UNABLE** to efficiently map SQL statements to appropriate functions.

2. **Demonstrations (10%)**
   SQL (DDL+DML)

Criteria	High (4)	Average (3)	Low (2)	Unsatisfied (1)
Appropriateness and Accuracy of Design	The student has created an **EXCELLENT** Database design that includes **ALL** or the **MAJORITY** (above 90%) of the following characteristics:	The student has created a **GOOD** Database design that includes **MOST** (above 50%) of the following characteristics: • At least 20 records have	The student has created an **ACCEPTABLE** Database design that includes **SOME** of the following characteristics: • At least 20 records	The student has created a **SIMPLISTIC** Database design that includes some of the following characteristics either superficially or is missing major components. The database is **NOT** suitable

Criteria	High (4)	Average (3)	Low (2)	Unsatisfied (1)
	• At least 20 records have been added to the Database • Appropriate Queries that include at LEAST FIVE (5) DML skills • Appropriate Report structure	been added to the Database • Appropriate Queries that include at LEAST FIVE (5) DML skills • Appropriate Report structure	have been added to the Database • Appropriate Queries that include at LEAST FIVE (5) DML skills • Appropriate Report structure **OR** There are misconceptions/mistakes within the data, query, or reports that significantly alter their appropriateness of them.	for the intended purpose.: • At least 20 records have been added to the Database • Appropriate Queries that include at LEAST TWO DML skills • Appropriate Report structure including.